

Arrays

1) How to create an Array?

Data Type variable name [Size];

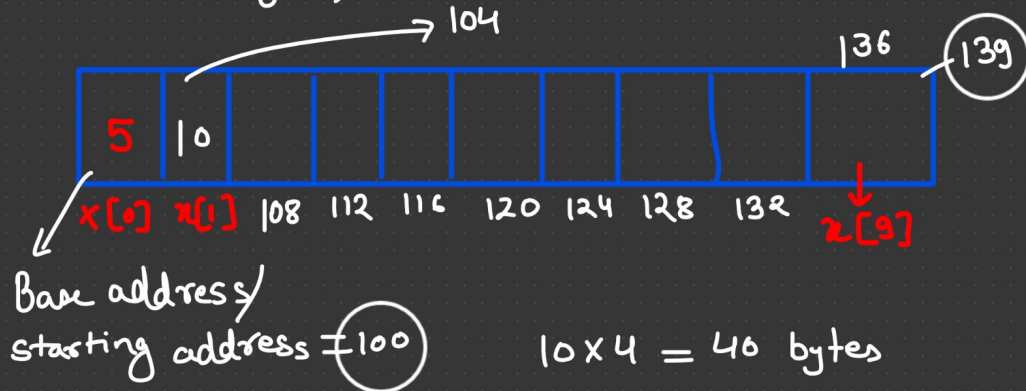
Ex:- `int x[10];`

`x[0] = 5;`

`x[1] = 10;`

⋮

`int = 4 byte`



≡ Array is collection of similar type of Elements, stored in contiguous memory allocation.

`int x[10];`
↓
Declaration

0	1	2	3	4	5	6	7	8	9
5	8	20	25	40	30	35	41	32	49
100	104	108	112	116	120	124	128	132	136

$X = \text{Base address} = 100$

$X[0] = \text{Value at } 100 = 5$

$X+1 = 104$

$X[1] = \text{value at } 104 = 8$

$X+6 = 124$

$X[6] = \text{value at } 124 = 35$

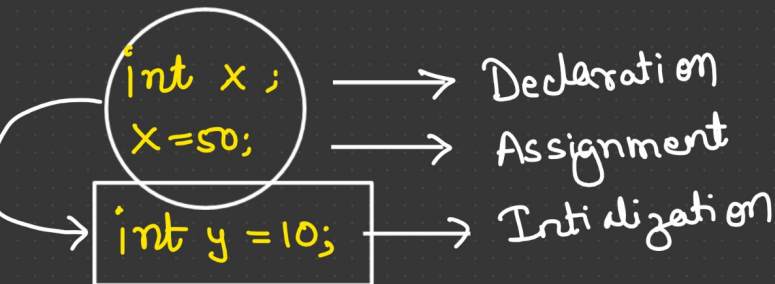
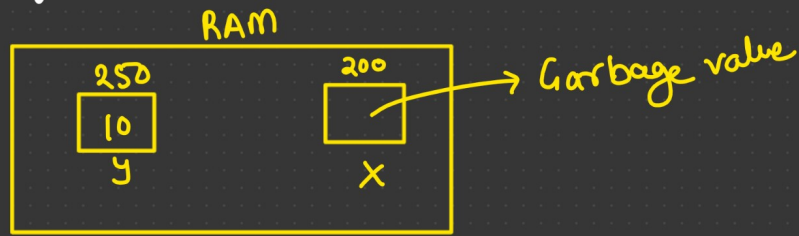
$x[0] = x + 0 \times 4 = 100 + 0 = 100$

$x[2] = 100 + 2 \times 4 = 108$

$x[6] = 100 + 6 \times 4 = 124$

$\text{Address} = \text{Base Address} + (\text{index} \times \text{size})$

Initialization of an Array:-



int a[5];
↓
Declare

1	2	3	4	5
0	1	2	3	4

int a[5] = {1, 2, 3, 4, 5}; // Initialization

int a[5] = {1, 2, 3}; // 1, 2, 3, 0, 0

int a[] = {1, 2, 3, 4, 5}; // size = 5.

2) Accessing Array Elements:-

```
for (i=0; i<n; i++)  
{  
    printf("%d", a[i]);  
}
```

i=0, a[0]=5

i=1, a[1]=10

i=2, a[2]=15

i=3, a[3]=25

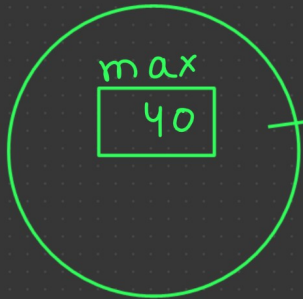
i=4, a[4]=35

5	10	15	25	35
0	1	2	3	4

Q:- WAP to find maximum element from the array.

↓
Write a program

20	18	10	15	30	25	35	28	32	40
0	1	2	3	4	5	6	7	8	9



```
max = a[0];  
for (i = 1; i < 10; i++)  
{  
    if (max < a[i])  
        max = a[i];  
}  
printf("Maximum is %d", max);
```

Q:- find the sum of all the elements in the given array.

1	4	5	3	4	7	6
0	1	2	3	4	5	6

Sum
x

$$1 + 4 = 5$$

$$5 + 5 = 10$$

$$10 + 3 = 13$$

$$13 + 4 = 17$$

Sum = 0;

```
for (i = 0; i < 7; i++)  
{  
    sum = sum + a[i];  
}
```

Q:- WAP to sort the given array in increasing order

{ 2, 6, 4, 9, 8 } $\longrightarrow$ { 2, 4, 6, 8, 9 }

a[0]

Swap(10, 6); | Swap(10, 6);
Swap(6, 4); | Swap(10, 8);

4	6	8	10	12	9	14
0	1	2	3	4	5	6

i=0 [1, 2, 3, 4, 5, 6]

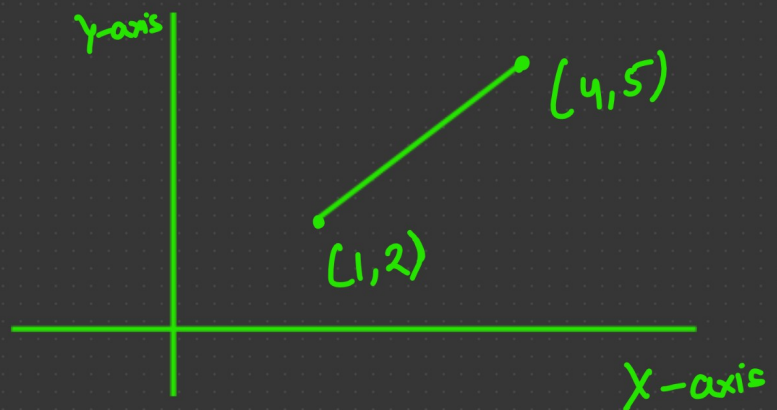
i=1 [2, 3, 4, 5, 6]

i=2 [3, 4, 5, 6]

```
for(i=0; i<6; i++)  
{  
    for(j=i+1; j<7; j++)  
    {  
        if(a[i] > a[j])  
            swap(a[i], a[j]);  
    }  
}
```

2D Array :-

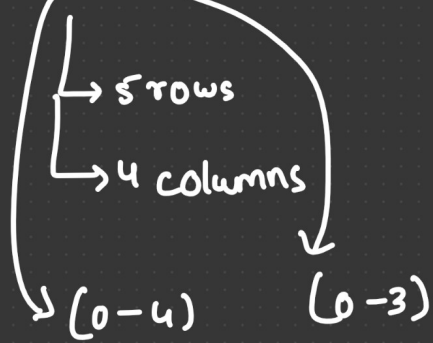
	0	1	2	3
Columns				
row $\rightarrow$ 0	00	01	02	03
$\rightarrow$ 1	10	11	12	13
$\rightarrow$ 2	20	21	22	23
$\rightarrow$ 3	30	31	32	33



Matrix = 4 $\times$ 4 = 16 cells
 $\downarrow$ $\downarrow$
 Row Column

int a[4][4];
 $\downarrow$ $\downarrow$
 Row Column
 (0-3) (0-3)

int a[5][4];



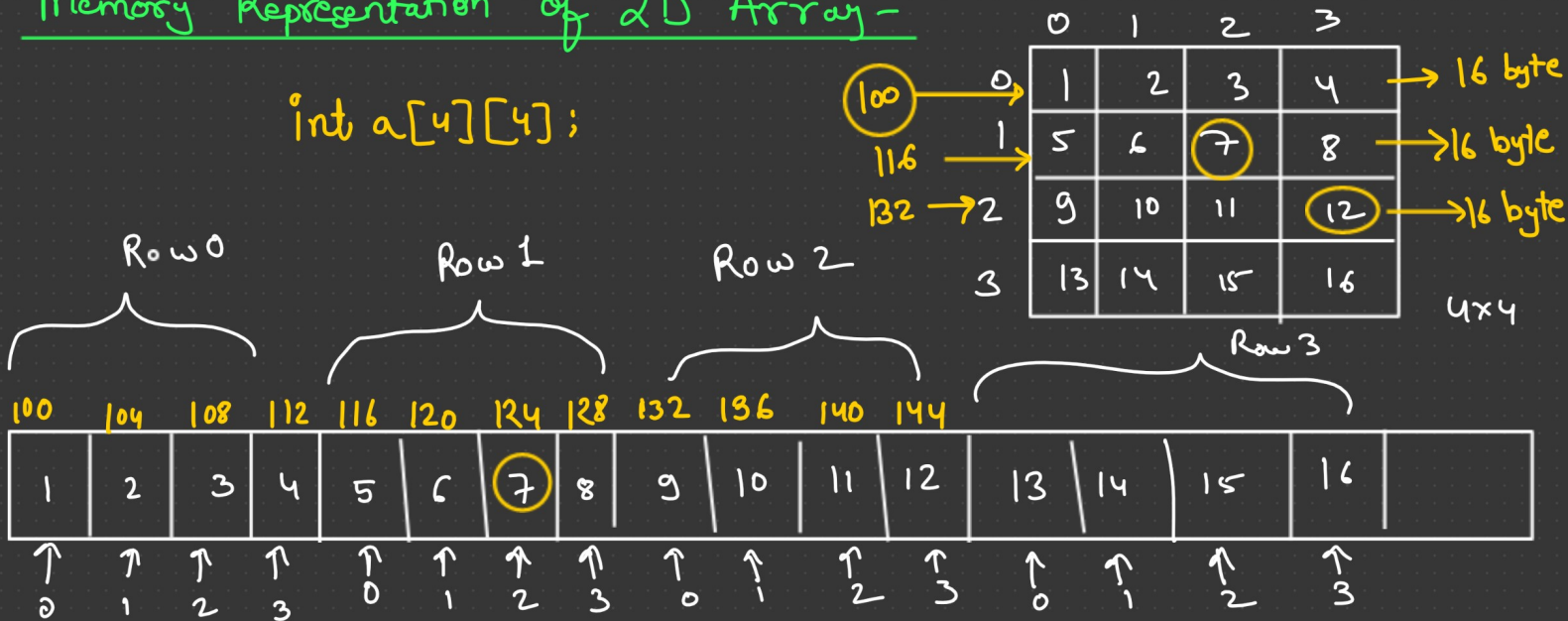
a[0][0] = 1;
a[0][2] = 4;
a[2][2] = 6;

	0	1	2	3	
0	0	1	2	3	← i
1	1	2	3	4	
2	2	3	4	5	
3	3	4	5	6	
4	4	5	6	7	↑ j

```
for (i = 0; i <= 4; i++) → i = 0, 1, 2, 3, 4
{
    for (j = 0; j <= 3; j++) → j = 0, 1, 2, 3
    {
        a[i][j] = i + j;
    }
}
```

Memory Representation of 2D Array -

int a[4][4];



$$a[1][2] = 100 + [(4 \times 4) \times 1] + (4 \times 2) = 124$$

$$a[2][3] = 100 + [(4 \times 4) \times 2] + (4 \times 3) = 144$$

(no. of column) size of (int)

int a[R][C];

a[i][j]

$$\text{Address} = \text{Base Address} + [C \times (\text{Size of Data Type}) \times i] + (\text{Size of DT}) \times j$$

Q:- Addition of two 3x3 matrix.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & -1 \\ 4 & 5 & 2 \end{bmatrix}_{3 \times 3}$$

$$B = \begin{bmatrix} 3 & 1 & 4 \\ 2 & 0 & 6 \\ -1 & 2 & 4 \end{bmatrix}_{3 \times 3}$$

$$A+B = \begin{bmatrix} 4 & 3 & 7 \\ 4 & 1 & 5 \\ 3 & 7 & 6 \end{bmatrix}$$

i=0 j=0,1,2

[0][0] [0][1] [0][2]

```
int a[3][3] = {1, 2, 3, 2, 1, -1, 4, 5, 2};  
int b[3][3] = {3, 1, 4, 2, 0, 6, -1, 2, 4};  
int c[3][3];
```

```
for (i=0; i<3; i++)
```

```
{ for (j=0; j<3; j++)
```

```
{ c[i][j] = a[i][j] + b[i][j];
```

```
}
```

```
}
```

Q:- Matrix multiplication 2×2 .

$$A = \begin{bmatrix} \textcircled{1} & \textcircled{2} \\ 3 & 4 \end{bmatrix} \begin{matrix} \rightarrow R_1 \\ \rightarrow R_2 \end{matrix}, \quad B = \begin{bmatrix} \textcircled{1} & \textcircled{-1} \\ \textcircled{2} & \textcircled{3} \end{bmatrix} \begin{matrix} \leftarrow C_1 \\ \leftarrow C_2 \end{matrix}$$

$$\begin{matrix} A & B \\ \begin{bmatrix} 00 & 01 \\ 10 & 11 \end{bmatrix} & \times \begin{bmatrix} 00 & 01 \\ 10 & 11 \end{bmatrix} \end{matrix}$$

$$A \times B = \begin{bmatrix} 1 \times 1 + 2 \times 2 & 1 \times (-1) + 2 \times 3 \\ 3 \times 1 + 4 \times 2 & 3 \times (-1) + 4 \times 3 \end{bmatrix}$$

$$\text{Res} \begin{bmatrix} 00 & 01 \\ 10 & 11 \end{bmatrix}$$

$$R=0 \times C1$$

$$R=0 \times C2$$

$$R=1 \times C1$$

$$R=1 \times C2$$

for ($i=0$) $i < 2$; $i++$ $\rightarrow i=0$

{ for ($j=0$) $j < 2$; $j++$ $\rightarrow j=0$

{ for ($k=0$) $k < 2$; $k++$ $\rightarrow k=(0, 1)$

$$i=0, j=0 \rightarrow \text{sum} = \begin{matrix} i & k & k & j \\ \begin{bmatrix} (00) \times (00) \\ (00) \times (01) \end{bmatrix} + \begin{bmatrix} (01) \times (10) \\ (01) \times (11) \end{bmatrix} \end{matrix}$$

$$c[i][j] = \text{sum};$$

{ }

$$\text{sum} = \text{sum} + a[i][k] * b[k][j]$$

$$\begin{matrix} i=1 \\ j=0, j=1 \\ k=(0,1) \end{matrix} \quad \left| \quad \begin{matrix} j=1 \\ k=(0,1) \end{matrix} \right. \quad \left| \quad \begin{matrix} k=(0,1) \end{matrix} \right.$$

$$a[i][k] * b[k][j] \rightarrow \text{sum}$$

A

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

B

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{aligned} (00) &= \underbrace{(00) \times (00)}_{A \quad B, K=0} + \underbrace{(01) \times (10)}_{A \quad B, K=1} \\ (10) &= \underbrace{(10) \times (00)}_{A \quad B, K=0} + \underbrace{(11) \times (10)}_{A \quad B, K=1} \end{aligned}$$

$$\begin{aligned} &\underbrace{(00) \times (01)}_{K=0} + \underbrace{(01) \times (11)}_{K=1} = (01) \\ &\underbrace{(10) \times (01)}_{K=0} + \underbrace{(11) \times (11)}_{K=1} = (11) \end{aligned}$$

```
for (i=0 ; i<2 ; i++)
{
```

```
    for (j=0 ; j<2 ; j++)
    {
```

```
        sum = 0;
```

```
        for (k=0 ; k<2 ; k++)
```

```
        {
            sum = a[i][k] * b[k][j] + sum;
```

```
        }
        c[i][j] = sum;
```

```
    }
```

```
}
```


Q:- Find Transpose of the given matrix?

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} \underline{00} & \underline{01} & \underline{02} \\ \underline{10} & \underline{11} & \underline{12} \\ \underline{20} & \underline{21} & \underline{22} \end{bmatrix} \rightarrow \begin{bmatrix} \underline{00} & \underline{10} & \underline{20} \\ \underline{01} & \underline{11} & \underline{21} \\ \underline{02} & \underline{12} & \underline{22} \end{bmatrix}$$

```
for (i=0; i<3; i++)
{
    for (j=0; j<3; j++)
    {
        b[i][j] = a[j][i];
    }
}
```

$$\begin{bmatrix} 00 & 01 & 02 \\ 10 & 11 & 12 \end{bmatrix} \quad 2 \times 3$$

$$\begin{bmatrix} 00 & 10 \\ 01 & 11 \\ 02 & 12 \end{bmatrix} \quad 3 \times 2$$

Q:- Sum of right diagonal / left diagonal?

$$\begin{bmatrix} 00 & 01 & 02 \\ 10 & 11 & 12 \\ 20 & 21 & 22 \end{bmatrix}$$

↓
Left

$$\begin{bmatrix} 00 & 01 & 02 \\ 10 & 11 & 12 \\ 20 & 21 & 22 \end{bmatrix}$$

↓
Right

```
for (i=0; i<3; i++)
{
    sum = a[i][i] + sum;
}
```

```
if (i+j==2)
    sum = a[i][j] + sum;
```

```
for (i=0, j=2; i<3; i++, j--)
    sum = a[i][j] + sum;
```

3D Array :-

DairyMilk = 0

DairyMilk = 1

No. of DM
↑
Row of DM0
Col of DM0
DM[0][1][2]

00	01	02	03
10	11	12	13
20	21	22	23

00	01	02	03
10	11	12	13
20	21	22	23

3x4

3x4

0 → (12)

1 → (12)

DM[1][1][2]

1	2	3	4
---	---	---	---

→ 4
→ 4
12

12 / 4 = 3

1 DM → 12 pieces (each row 4 piece)

Box → 4 DM

Size of Box = ?

Box[4][3][4];
no. of DM no. of Rows no. of Col
4x3x4

int a[3][4][2];

	0	1
0		
1		
2		
3		

	0	1
0		
1		
2		3
3		

	0	1
0		
1		
2		
3		

No. of element

= 3x4x2

= 24

a[1][2][1] = 3;

`int a[2][2][3] = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 };`

`int a[2][2][3] = { { { 1, 2, 3 }, { 4, 5, 6 } }, { { 7, 8, 9 }, { 10, 11, 12 } } };`

1	2	3
4	5	6

0

7	8	9
10	11	12

1

`int a[ ][2][3] = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 };`
 0 0 1 1
 1 matrix 2 matrix

`int a[ ][3][2] = { 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17 };`
 0 0 0 1 1 1 2 2 2
 ① ② ③

`for(i=0; i<3; i++)`

`{ for(j=0; j<3; j++)`

`{ for(k=0; k<2; k++)`

`{ a[i][j][k] = i+j+k;`

`;`

`}`

`}`

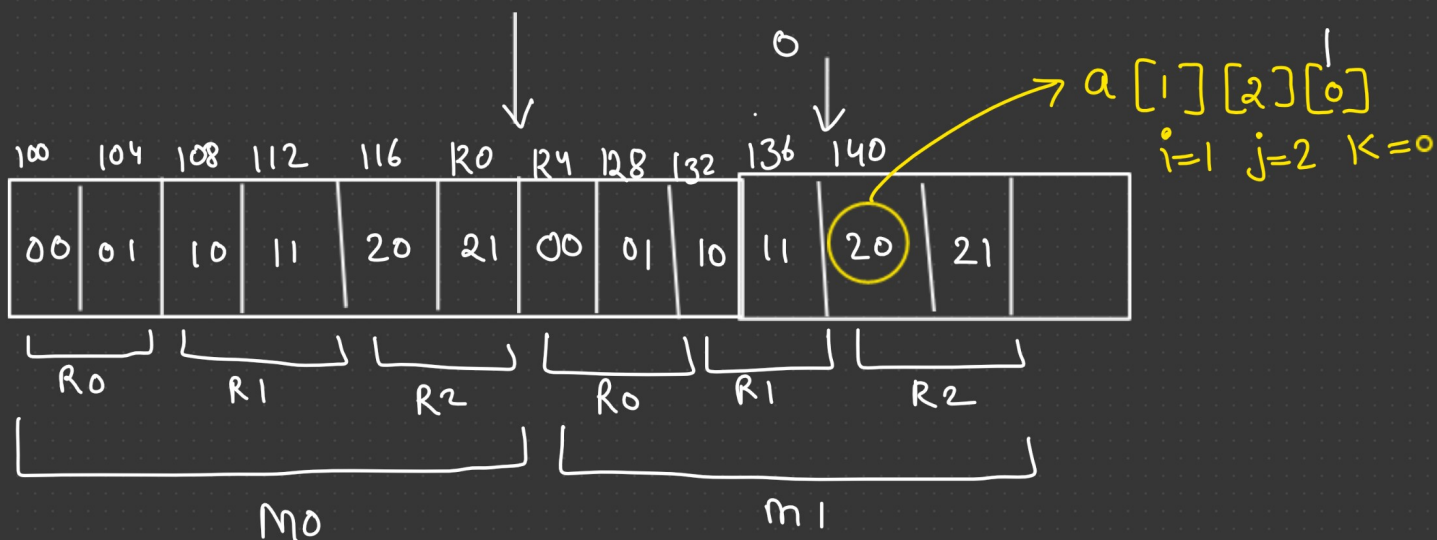
int a[2][3][2];

N R C

100

00	01
10	11
20	21

00	01
10	11
20	21



$$a[1][2][0] = 100 + [1 \times (3 \times 2 \times 4)] + [2 \times (2 \times 4)] + [0 \times 4]$$

$$= 100 + 24 + 16 + 0 = 140$$

$$\text{Address} = \text{Base Address} + [i \times (R \times C \times \text{size})] + [j \times (C \times \text{size})] + (k \times \text{size})$$

Q:- WAP to check the given matrix / Det. is sparse or not?

$$\begin{bmatrix} - & 0 & - \\ - & - & 0 \\ - & - & 0 \end{bmatrix}$$

$3 \times 3 = 9$ elements

$$\frac{9}{2} = 4.5 \leq \text{zero's} \Rightarrow \text{sparse}$$

$$\Rightarrow \text{dense}$$


```

for(i=0; i<n; i++)
{
    for(j=0; j<m; j++)
    {
        if(a[i][j] == 0)
            count++;
    }
}

if(count > (m*n)/2)
    printf("sparse");
else
    printf("dense");

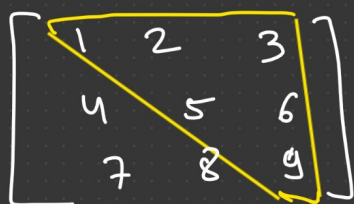
```

```

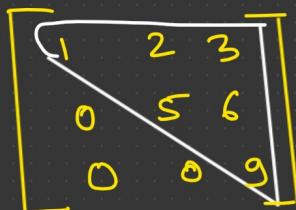
int a[n][m];
int count=0;

```

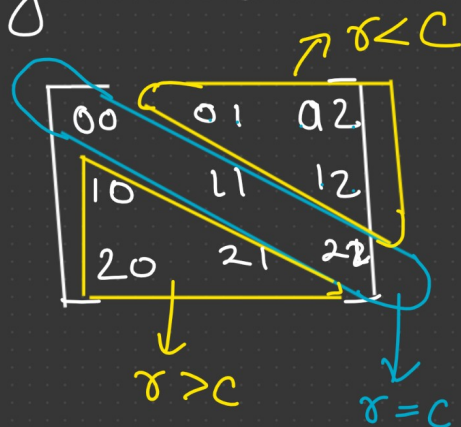
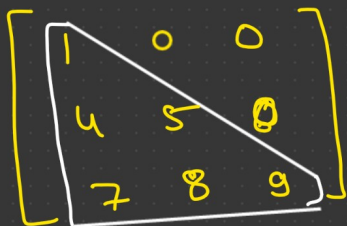
Q. WAP to print lower & upper triangular matrix:-



⇒



⇓



```

for(i=0; i<n; i++)
{
    for(j=0; j<m; j++)
    {
        if(i <= j)
            printf("%d", a[i][j]);
        else
            printf("0");
    }
}

```

WAP to find 2nd largest element in the given array?

1	4	10	8	7	9	5	3	2	11
---	---	----	---	---	---	---	---	---	----

max1 = ~~1~~ ~~4~~ ~~10~~ 11

max2 = ~~2~~ ~~3~~ ~~5~~ ~~7~~ ~~8~~ 9 10

```
for(i=0; i<n; i++)  
{  
    if(max1 < num)  
    {  
        max2 = max1;  
        max1 = num;  
    }  
    else if(max2 < num)  
        max2 = num;  
}
```

```
printf("Second Largest no. is %d", max2);
```

Array & functions

main()

{

f1(); // TNRN

f2(n); // TSRN

int y = f3(); // TNRS

int z = f4(y); // TGRS

}

main()

{

int a = 5, b = 10;

add(a, b);

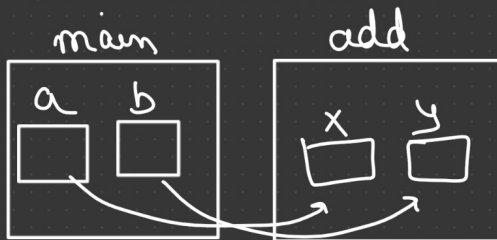
==

}

void add(int x, int y)

{

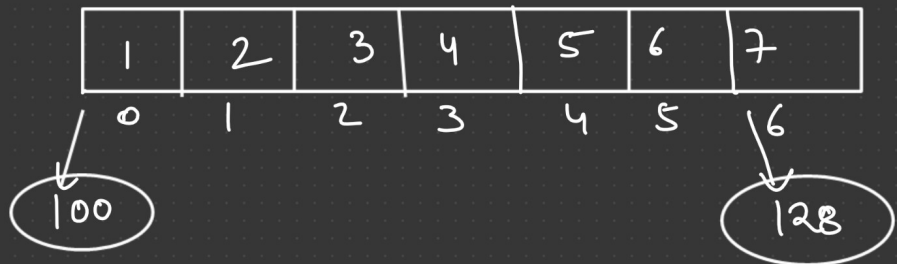
}



int a[7];

a[0] = 1;

a[2] = 3;



a = &a[0] = $\text{base address} + \text{Data type} + \text{no. of element}$

(int) 4-byte = 100, 101, 102, 103

$$100 + 7 \times 4 = 128$$

```

main( )
{
    int a[10];
    fun( a , 10)
}

fun( int a[ ], int size)
{
    for(i=0; i<size; i++)
        a[i];
}

```

Diagram illustrating function call and parameter passing:

- A vertical line separates the `main` function from the `fun` function.
- An arrow points from the variable `a` in `main` to the parameter `a` in `fun`.
- An arrow points from the value `10` in `main` to the parameter `size` in `fun`.
- Yellow annotations:
 - `a` in `main` is labeled "Base address".
 - `10` in `main` is labeled "num".

WAP to find largest number in a given array using fm.

```

main()
{
    int a[5] = { 2, 4, 3, 5, 1 };
    findMax(a);
}

```

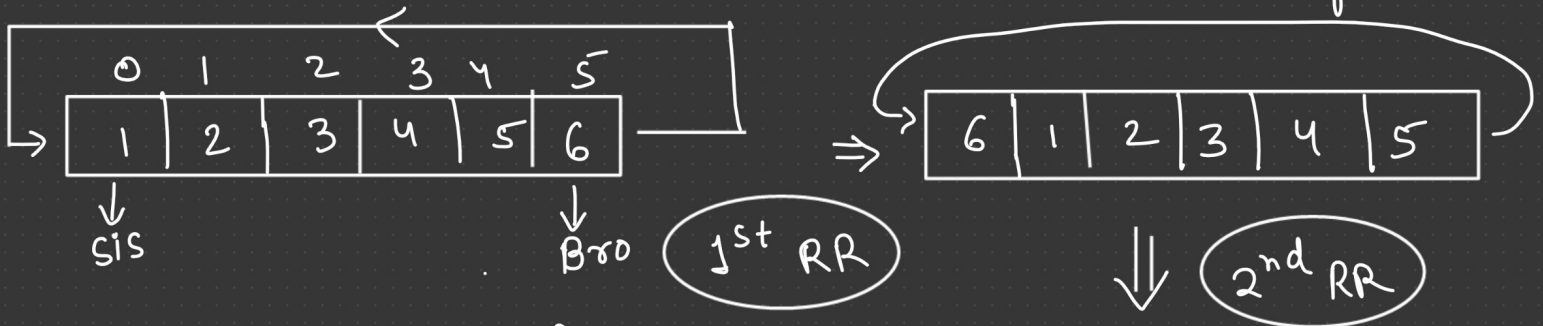
```

Void findMax( int x[ ])
{
    int i, max = 0;
    for(i=0; i<5; i++)
    {
        if(max < x[i])
            max = x[i];
    }
    printf("Max %d", max);
}

```


WAP to print all the elements of a given array using function.

WAP to rotate right by "d" distance in a given array of length "n".



```
for (j = 1; j <= d; j++)
```

```
{ int x = a[size - 1];
```

```
  for (i = size - 1; i > 0; i--)
```

```
    a[i] = a[i - 1];
```

```
  }
```

```
    a[0] = x;
```

```
}
```



WAP to find frequency of all the no. in a given array.

1	4	5	3	4	1	2	5	3	4	5
---	---	---	---	---	---	---	---	---	---	---

No.	frequency
1	→ 2 times
2	→ 1
3	→ 2
4	→ 3
5	→ 3

Count = ~~1~~ 2

1	4	5	3	
---	---	---	---	--

visited

1, 1	2	3, 3	4, 4, 4	5, 5, 5
------	---	------	---------	---------

```
for(i=0; i<n; i++)
{
    for(j=0; j<n; j++)
    {
        → n x n times
    }
}
```

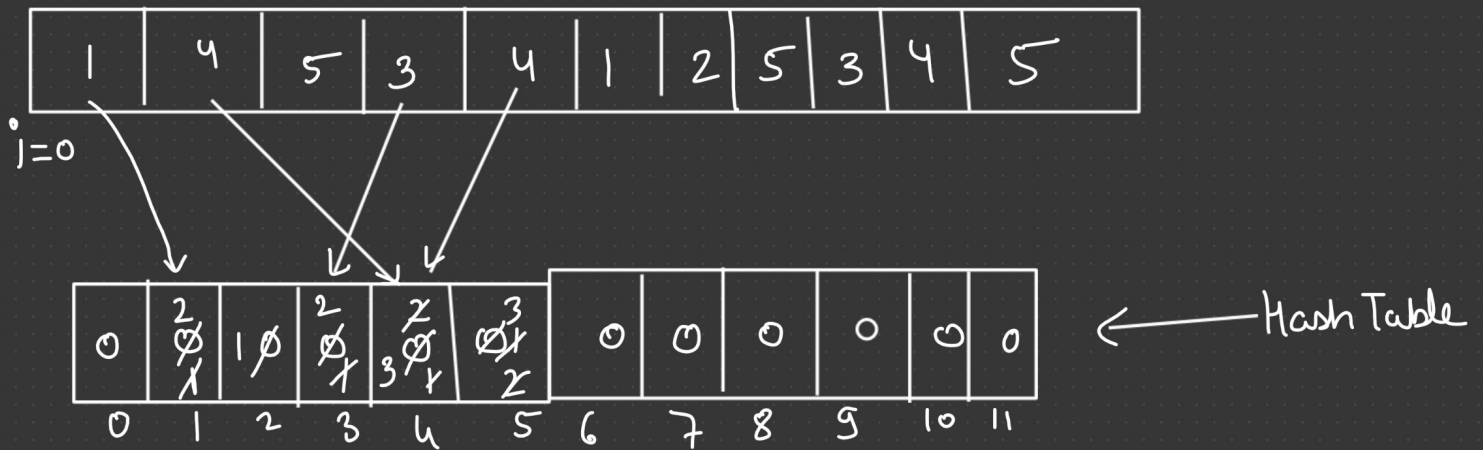
$n^2 > 2n$

```
for(i=0; i<n; i++)
{
    → n
}
```

```
for(j=0; j<n; j++) +
{
    → n
}
2n
```

↑
This code is same as one loop code.

Hashing Technique



1 → 2

2 → 1

3 → 2

4 → 3

5 → 2

```
int hash[100];  
  
for(i=0 ; i < size; i++)  
{  
    hash[a[i]]++;  
}
```

a[i]

Q:- WAP to find duplicate number/numbers?

```
main()
{
    int a[5][10] = { . . . };

    fun(a, 5, 10);

}

fun(int a[][10], int i, int j)
{
}
```

String in C

Sequence of characters is called string.

'P', 'R', 'A', 'T', 'E', 'E', 'K' $\Rightarrow$ characters

"prateek" $\Rightarrow$ string.

Declaration of String :-

char name[] = { 'P', 'R', 'A', 'T', 'E', 'E', 'K', '\0' } ;

null character



predefined fn $\rightarrow$ string $\rightarrow$ printf

'P' $\rightarrow$ character
"P" $\rightarrow$ string

printf(" %s", name);

%d, %f, %C

2) `char name[6] = {'A', 'B', 'C', 'D', 'E', '\0'};`

3) `char name[] = "Prateek";`

Input:-

1) `scanf("%s", a);`

2) `gets(a);`

3) `fgets(a, max, stdin);`

↳ `a[strlen(a)-1] = '\0';` → v. Imp

WAP to count number of vowels in a given string?

ex:-

"Prateek" → 3 (a, e, e)

WAP to convert uppercase to lowercase & vice versa.

PrATeeK → pRatEEK

'a' = 97
- 'A' = 65

32

'a' - 32 = 'A'
'A' + 32 = 'a'

'z' = 122
'Z' = 90

WAP to check the given string is palindrome or not.

"aabbbaa" ✓
↑ ↑ ↑ ↑
i=0 j=size-1

→ ←
i++ j--

"Nayan" ✓
→ ←

WAP to find frequency of all character in the given string.

Ex:- char a[] = "ababccac";

a = 3
b = 2
c = 2
int hash[200] = {0};

```
for(i=0; a[i] != '\0'; i++)
```

```
{
```

```
    hash[a[i]]++;
```

```
}
```

```
for(i=0; i<123; i++)
```

```
{ if(hash[i] > 0)
```

```
    printf("%c = %d", i, hash[i]);
```

```
}
```

Multiple Strings

char a[] = "Prateek";

P	R	A	T	E	E	K	\0		
a	0	1	2	3	4	5	6	7	8

a[0] = 'P'

a[5] = 'E'

a[3] = 'T'

char a[5][10] = { "Prateek", "Shubham", "Milon", "Arnab", "Prashant" };

String No. (R)
Char no. of that string. (C)

a[0] = "Prateek";

a[1] = "Shubham";

a[2][3] = 'a'

a[3][2] = 'n'

	0	1	2	3	4	5	6	7	8	9
a[0] → 0	p	r	a	t	e	e	k			
a[1] → 1	s	h	u	b	h	a	m			
a[2] → 2	m	i	l	a	n					
a[3] → 3	A	r	n	a	b					
a[4] → 4	p	r	a	s	h	a	n	t		

```
for(i=0; i<5; i++)  
    printf("%s\n", a[i]);
```

```
for(i=0; i<5; i++)  
{  
    for(j=0; a[i][j] != '\0'; j++)  
    {  
        if(a[i][j] == 'a')  
            count++;  
    }  
}
```

WAP to sort given names in dictionary order.

- 1) strcmp("Prateek", "Prashant"); > 0 (swap)
- 2) strcmp("Prateek", "Prateek"); $= 0$ (equal)
- 3) strcmp("Prashant", "Prateek"); < 0 (no need to swap)

```
char name[5][20];  
char temp[20];
```

$n=5$

```
for(i=0; i < n-1; i++)  
{
```

```
    for(j=i+1; j < n; j++)  
    {
```

```
        if(strcmp(name[i], name[j]) > 0)
```

```
        {  
            strcpy(temp, name[i]);  
            strcpy(name[i], name[j]);  
            strcpy(name[j], temp);
```

```
        }
```

```
    }
```

```
}
```

"abc" $\rightarrow 0$
"adc" $\rightarrow 1$
"bca" $\rightarrow 2$

$n=3$

	0	1	2	3	4	5	6	7	8	9
a[0] $\rightarrow 0$	A	r	n	a	b					
a[1] $\rightarrow 1$	s	h	u	b	h	a	m			
a[2] $\rightarrow 2$	P	r	a	t	e	e	k			
a[3] $\rightarrow 3$	M	i	l	a	n					
a[4] $\rightarrow 4$	p	r	a	s	h	a	n	t		

WAP to search a name in the given list of Strings.

```
char a[5][10] = { "Prateek", "Shubham", "Milon", "Arnob", "Prashant" };
```

```
char name[] = "milon";
```

```
for (i = 0 ; i < 5 ; i++)
```

```
{  
    if ( strcmp( name, a[i] ) == 0 )
```

```
        printf("String found");
```

```
}
```